Lab Two – Basic Tasks in R

Use `R` (R Core Team 2025) to complete the tasks below.

1 Libraries and Packages

1.1 Part a

Install the `esquisse` package for `R`. Report the code below, ensuring to set the `eval` option to `false`, using "#| eval: false", so that we don't install the package every time you compile your Quarto document.

Solution

```r
install.packages ("esquisse")
```

1.2 Part b

Load the `esquisse` package for `R`. Report the code below. It does not matter whether you set "#| eval: false" because loading the package does not add a lot of time. Note you will not evaluate any code that uses the `esquisse` package, so there is no need to load it as you compile your Quarto document.

Solution

```r
library ("esquisse")
```

1.3 Part c

How can you ask for help about the `esquisse` package for `R`? Report the code below, ensuring to set the `eval` option to `false`, using "#| eval: false", so that the help document isn't loaded every time you compile your Quarto document.

Solution

```r
help ("esquisse")
```

1.4 Part d

Is a demo or vignette available for the `esquisse` package for `R`? Report the code below, ensuring to set the `eval` option to `false`, using "#| eval: false", so that the you don't get errors when you compile your Quarto document.

Solution

No demo was found for the package. The following vignettes were found: `get-started` and `shiny-usage`.

```r
demo ("esquisse")
```

```r
vignette ("esquisse")
```

1.5 Part e

Add the BibTeX citation for `esquisse` package for `R` to your `.bib` file and cite it in a sentence describing what the package does below. Note you can reference a citation named `esquisse` using "[@esquisse]".

Solution

We use `esquisse` (Meyer and Perrier 2025) to explore and visualize data interactively.

1.6 Part f

Run `esquisser()` in your console – not in a chunk of `R` code in your Quarto document.

i. Select `palmerpenguins` from the "Select an environment in which to search:" dropdown. This should automatically select `penguins` in the "Select a data.frame:" dropdown. Click "Import Data".
ii. Drag `body_mass_g` to the X box and `species` to the fill box.
iii. Describe what you see in words. What can you conclude about Adelie, Chinstrap, and Gentoo penguins based on the resulting plot?

Solution

Looking at the 3 boxplots, we observe that the median body masses of Chinstrap and Adelie penguins are lower than that of Gentoo penguins. The body mass of Gentoo penguines have greater variability than the other groups'. In the dataset for the Chinstrap group, there are 2 outliers.

2 Objects and Vectors

Create the following vectors in `R`. In some cases, you may be able to use `seq()` or `rep()` while in others you cannot. Use these functions when possible, otherwise manually create the vector and explain why that was necessary.

2.1 Part a

The Fibonacci Sequence is a recursive formula:

$$F_n = F_{n-1} + F_{n-2}$$

where $F_0 = 0$ and $F_1 = 1$.

Create a vector of the first 10 Fibonacci numbers.

Solution

`seq(...)` cannot be used because the distance between 2 observations in these vectors are not fixed. The numbers in this sequence are also not repeated, so I can't use `rep(...)`.

```r
(fibonacci <- c(0, 1, 1, 2, 3, 5, 8, 13, 21, 34))
```

```
 [1]  0  1  1  2  3  5  8 13 21 34
```

2.2 Part b

Triangular Numbers are the cumulative sums of natural numbers:

$$F_n = \frac{n(n+1)}{2}.$$

Create a vector of the first 10 Triangular Numbers.

Solution

We can use vectorized math to create this vectore

```r
n <- 1:10
triangular <- (n*(n+1))/2
(triangular)
```

```
 [1]  1  3  6 10 15 21 28 36 45 55
```

2.3 Part c

Suppose I were designing a repeated meausres experiment with three treatment conditions. Each of $n = 10$ participants (with `ID` 1 to 10) will receive *all* experimental conditions, call them "Control", "Treatment A", and "Treatment B".

Consider setting up data entry for this experiment.

i. Create a vector containing each `ID` repeated three times, once for each treatment.
ii. Create a vector containing each `Treatment` repeated for each participant ID.

Solution

```
(idrepeated <- rep(1:10, each=3))
```

```
 [1]  1  1  1  2  2  2  3  3  3  4  4  4  5  5  5  6  6  6  7  7  7  8  8  8  9
[26]  9  9 10 10 10
```

```
(txrepeated <- rep(c("Control", "Treatment A", "Treatment B"), times=10))
```

```
 [1] "Control"     "Treatment A" "Treatment B" "Control"     "Treatment A"
 [6] "Treatment B" "Control"     "Treatment A" "Treatment B" "Control"
[11] "Treatment A" "Treatment B" "Control"     "Treatment A" "Treatment B"
[16] "Control"     "Treatment A" "Treatment B" "Control"     "Treatment A"
[21] "Treatment B" "Control"     "Treatment A" "Treatment B" "Control"
[26] "Treatment A" "Treatment B" "Control"     "Treatment A" "Treatment B"
```

2.4 Part d

Create a vector containing the `character` "MATH" and the `numeric` 240. What is the resulting class? Explain why in a sentence.

Solution

Create a vector containing the `character` "MATH" and the `numeric` 240. What is the resulting class? Explain why in a sentence.

Solution

The resulting class is `character`. Vectors in R can only hold one type of data, so when types conflict R coerces everything to the most flexible type. Since any number can be represented as a string but not every string can be represented as a number, the `numeric` 240 becomes the `character` "240".

References

Meyer, Fanny, and Victor Perrier. 2025. *Esquisse: Explore and Visualize Your Data Interactively*. https://doi.org/10.32614/CRAN.package.esquisse.

R Core Team. 2025. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. https://www.R-project.org/.